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Chapter 10

Facing The Nemesis: Game Theory

To this point, we have explored various methods and examples of exploiting weaknesses in our opponent's strategies that are weak in one way or another. It may be the case that all your opponents are ripe for exploitation and will never adapt or attempt to exploit you in turn. If this is the case, then exploitive play will serve to generate profit. However, many games contain a mixture of strong and weak players; and in a related vein, tournaments often preclude game selection decisions by providing players with no choice of opponents. In the extreme case, you may actually find yourself in a game with players stronger than you are.

In these circumstances, what can you do? One approach is to continue to try to exploit the opponent's mistakes, playing a regular exploitive game and effectively concede an edge to the opponent. This is the approach favored, or perhaps we should say used, by many players, often for lack of a better alternative.

And a related situation occurs fairly frequently - that is, playing against new opposition with whom we are not familiar. One approach to playing in this situation is to simply play a "default" exploitive game, assuming the new player meets a typical profile and playing accordingly, while attempting to gain information about his play and adjusting as indicated. Again, this is the approach favored by the majority of players, often because they lack another reasonable alternative.

We, however, advocate a different approach in both of these circumstances. Instead of relying solely on an exploitive strategy, we attempt to find strategies that are fundamentally sound; that yield no benefit to an opponent's attempts to exploit us. We may in this way give up a small amount of potential profit that we could have seized by playing exploitively. But by doing this, we attempt to make ourselves in turn unexploitable.

This approach to the game finds its roots in game theory, a subject that occupies the remainder of Part III. In Part II, we introduced the concepts of ***games*** and ***strategies,*** which are the main topics with which game theory is concerned. This branch of mathematics is a large and growing field with applications in a broad array of disciplines. However, since this is a poker book, we will not attempt to provide a comprehensive overview of the many interesting situations game theory addresses except for those that lend insight into poker. Instead, we will introduce the various concepts of game theory as they are needed for our discussion, without attempting to be rigorously formal.

In part II, we defined a game as containing the following elements:

- There are two or more players.
- At least one player has a choice of actions.
- The game has a set of outcomes for each player.
- The outcomes depend on the choices of actions by the players.

We call the choice of actions a ***strategy,*** and the outcomes ***payoffs.*** Each action that a player may take is called a ***strategic option;*** the specification of all the choices he makes at each possible decision point is the strategy.

As a simple introduction to the structure of games, let us consider one of the simplest games, Odds and Evens. Two players play. Each conceals either a penny or no penny in his hand, and the hands are revealed simultaneously. By prior agreement, one player (Player A) win 1 unit if

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the total number of pennies in the hands is even, and the other player (Player B) wins 1 unit if the total number of pennies is odd.

The following is what is called a ***payoff matrix*** for this simple game. Player B's strategies are listed left to right and Player A's from top to bottom. Once the strategies have been revealed, looking up their intersection in the chart gives the payoff due to each player (A's payoff, B's payoff).

Example 10.1 - Odds and Evens

	Player B	
Player A	0 pennies	1 penny
0 pennies	(1, - 1)	(-1, 1)
1 penny	(-1, 1)	(1, - 1)

There are a number of properties of interest that games can have. Some games, such as Odds and Evens or rake-free poker, are what are called ***constant-sum*** games. The total of all the payoffs to the various players must total a constant. In poker, this constant happens to be zero, and it is common to speak of ***zero-sum*** games as well. All constant-sum games can be converted to zero sums by simply scaling the payoffs appropriately.

A second important property of games is whether the play is ***sequential or simultaneous.*** In a game like Odds and Evens, for example, the play is simultaneous. In a game like checkers, for example, play is sequential; first one player plays, then the other. In some sense, we can convert all games to simultaneous play by simply requiring all players to specify their complete strategy (including all possible game trees) beforehand. However, the distinction between simultaneous and sequential play is still useful because of what it tells us about mixed strategies (more in a moment).

Another important property of games is whether there is ***hidden information.*** Hidden information is information available to one player but not the other. First consider a game such as chess, hi this game, the entire game situation is available to either player at any time, and the game is completely deterministic; that is, a player or computer with enough calculating power could examine every possible game tree and find the best move. This is an example of a game with no hidden information. Backgammon is an example of a game where there is no hidden information, but the game is not deterministic. On each roll of the dice the game tree forks, but the mechanics of this probabilistic selection process are known to each player. So in this case, even though perfect play cannot guarantee victory for either player because of the random element, there is no hidden information; that is, there is no information that one player has that the other does not.

Poker, on the other hand, is a game that relies on hidden information. Each player knows his own hand, and the knowledge of his opponent's hands would be a great boon to him. As a result, the task of poker is to extract value by taking advantage of the hidden information. From a game theory standpoint, the most important thing about games with hidden information is that their optimal strategies can contain ***mixed strategies.*** Mixed strategies are situations where strategies contain two or more actions in a single situation. For example, "raise half the time, limp- reraise half the time with aces in a full game" is an example of a mixed strategy. Only games with either hidden information or sequential play can contain mixed optimal strategies.

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In using game theory to study poker, ***zero-sum two-player games*** (such as Odds and Evens) are normally the most important and most interesting. In zero-sum two-player games, only two players have payoffs and those payoffs always sum to zero; that is, whatever player A loses, player B wins, and vice versa. The comments that Follow about "optimal strategies" apply only to zero-sum two-player games.

We can discuss ***strategy pairs;*** that is, a combination of a strategy for the first player and a strategy for the second player. For a zero-sum two-player game, a strategy pair is ***optimal*** if neither player can improve his expectation by unilaterally changing his strategy. A strategy for one player is optimal if it is part of any optimal strategy pair.

Imagine playing against a super-opponent. We call this super-opponent the ***nemesis***. The nemesis always knows your strategy and always plays the maximally exploitive strategy against it. If you change your strategy, the nemesis changes instantly to counter, always playing the strategy that maximally exploits yours. An optimal strategy is the strategy that has maximum EV against the nemesis. Another way of stating this is:

An optimal strategy pair consists of two strategies that maximally exploit each other.

Please note that the term "optimal" in this context is carefully defined - many disciplines use this term, sometimes with varying definitions or meaning. We will use the term only in the narrow sense given above and only in reference to strategies.

In both zero-sum two-player and other types of games, either variable-sum or multi-player, we can still satisfy the condition that no player be able to increase his expectation by acting unilaterally. Strategy sets for each player that satisify this condition are called ***Nash equilibria.*** It is proven that all multiplayer games with finite payout matrices have at least one such equilibrium; some games have multiple equilibria. In fact, some games have multiple equilibria that makes them difficult to analyze. We will revisit this In Part V.

Mathematics tells us that the optimal strategies to a ***zero-sum two-player game*** have the following properties:

- As long as mixed strategies are allowed (i.e., each player can use strategies such as "do X 60% of the time and Y 40%"), optimal strategies always exist.
- As a corollary to this, if an optimal strategy contains a mixed strategy, then the expectation of each strategic alternative must be equal against the opponent's optimal strategy. Thus, optimal strategies in poker do not contain "loss leaders" or other plays that sacrifice immediate expectation for deceptive purposes. If a hand is played in different ways, then each way of playing the hand will have the same expectation. If this were not the case, then the player could simply shift all the hands from the option with lower expectation to the option with higher expectation and unilaterally improve.

Frequently, especially in very simple games, the optimal strategy is simply the one that guarantees zero EV. This is because simple games are frequently totally symmetric; in this case the optimal strategy must yield zero for the player employing it, as his opponent could simply play the optimal strategy in response.

Looking back at Odds and Evens, it is clear that Player A's objective is to match Player B's - strategy. The two could play this game as an exploitive guessing game and try to outthink each other; in that case, the player better at outguessing his opponent would have an edge. Another option is available, however. Suppose that B felt he was inferior at the guessing game to A. B

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could instead try to play optimally. One way of doing this is to try to find the nemesis srategy for any strategy he himself utilizes and then maximize his equity.

B can play any mixed strategy which consists of playing 0 pennies X% of the time and 1 penny (1-X)% of the time. Note that pure strategies are simply a mixed strategy where one of the options is given 100% weight.

We can directly calculate the expectation of A's counter-strategies, but we should know from our work on exploitive play that the nemesis will best exploit B by playing a pure strategy. If B plays 0 pennies more than 50%, the nemesis will play 0 pennies all the time. If B plays 1 penny more than 50%, the nemesis will play 1 penny all the time.

From Equation 1.11, B's EV for 0 pennies > 0.5 is:

<B, x > 0.5 > = (-l)(x) + (1)(1-x)
<B, x > 0.5 > = 1 – 2x

His EV for 1 penny > 0.5 is:

<B, x < 0.5 > = (-1)(1 - x) + (l)(x)
<B, *x* < 0.5 > = 2x -1

We can see that in both of these cases, the expectation of the strategy is negative.
When x > 0.5,1- 2x is negative, while when x < 0.5, 2x - 1 is negative.

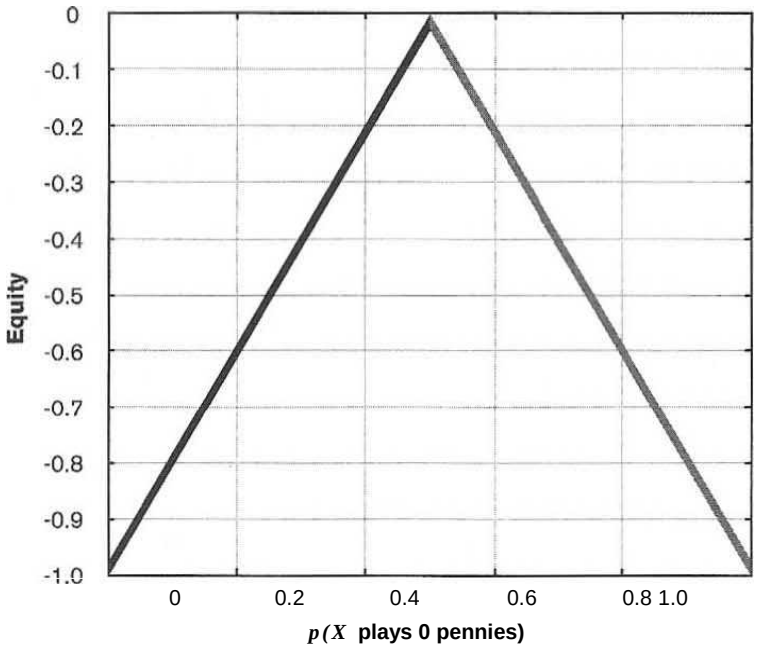

Figure 10.1. Odds and Evens equity vs. nemesis

At precisely ***x*=** 0.5, the nemesis can do anything and achieve the same equity.

<nemesis, 0 pennies > = (-1)(0.5) + (1)(0.5)
<nemesis, 0 pennies > = 0
<nemesis, 1 penny > = (-1)(0.5) + (1)(0.5) = 0
<nemesis, 1 penny > = 0

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This is the highest expectation B can have against the nemesis. Therefore, $x = 0.5$ is B's optimal strategy. B can guarantee himself zero expectation by following a strategy where he randomly selects between 0 pennies (50%) and 1 penny (50%). To do this, he could use any source of external randomness; a die roll, another coin flip, or perhaps quantum measurements ■*i* radioactive decay, if he were sufficiently motivated to be unexploitable.

Example 10.2 - Roshambo

A second, slightly more complicated game that illustrates the same concept is the game of Roshambo (also known as Rock, Paper, Scissors). In this game, the players choose among three options: rock, paper, or scission, and the payoff matrix is as follows:

	Player B		
Player A	Rock	Paper	Scissors
Rock	(0, 0)	(-1, 1)	(1, -1)
Paper	(1, -1)	(0, 0)	(-1, 1)
Scissors	(-1,1)	(1, -1)	(0, 0)

In general, throughout this book we will be referring to zero-sum games. In cases where we have games that are non-zero sum, we will denote it explicitly. The following matrix is equivalent to the above, except that only A's outcomes are in the matrix cells.

	Player B		
Player A	Rock	Paper	Scissors
Rock	0	-1	+1
Paper	+1	0	-1
Scissors	-1	+1	0

If B wants to ensure 0 EV, he can play the strategy $\{1/3, 1/3, 1/3\}$ a similar way to the calculation in Odds and Evens, no matter what option A chooses, he will win 1 unit $1/3$ of the time, tie $1/3$ of the time, and lose 1 unit *Yi* of the time. Any other choice on the part of B will lead to the nemesis exploiting him by playing a pure strategy in response. So this is an optimal strategy for this game.

What occurs frequently in these types of games is a situation where the nemesis has two or more possible exploitive strategies to employ. B selects his strategy, and the nemesis selects his exploitive strategy in response. At some strategy S that B chooses, the nemesis strategy is ***indifferent*** between his exploitive strategy choices; that is, it makes no difference to his expectation which one he chooses. This ***indifference threshold*** is the optimal strategy for B. The reason that this concept is so powerful is that it enables us to ensure that the strategies we find cannot be improved by moving them a tiny bit in a different direction. Remember that one of the conditions for a strategy being optimal is that neither player be able to improve by changing his strategy unilaterally. When a player is indifferent between strategy choices, this prevents this condition from being violated.

To see how this allows us to more easily find optimal strategies, let us consider a slightly modified version of Roshambo, where there is a bonus for winning with scissors. We'll call this game Roshambo-S, and its payoff matrix is as follows:

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Example 10.3 - Roshambo - S

	Player B		
Player A	Rock	Paper	Scissors
Rock	0	-1	+1
Paper	+1	0	-2
Scissors	-1	+2	0

Now the strategy $\{1/3, 1/3, 1/3\}$ is easily foiled by the opponent playing {0, 0, 1}, in which case the opponent wins $1/3$ of a unit overall. However, A can again reduce the game to 0 EV by attempting to make his opponent indifferent between the various choices.

Say A's strategy will be *{a, b, c}*. Then A wants to make B's EV from the various strategy options equal. Using Equation 1.11:

<B, rock > = (0)(a) + (-l)(b) + (1)(c)
<B, rock > = c - b
<B, paper > = (1)(a) + (0)(b) + (-2)(c)
<B, paper > = a - 2c
<B, scissors > = (-1)(a) + (2)(b) + (0)(c)
<B, scissors > = 2b - a

Setting these three things equal and solving the resultant system of equations, we find that $a = 2b=2c$. This indicates a strategy of $\{1/2, 1/4, 1/4\}$. We can verify that this strategy is optimal indirectly. First of all, the game is symmetrical, so any strategy B can utilize, A can utilize also. Since the payoffs sum to zero, no strategy can have positive expectation against the nemesis. If such a strategy existed, then both players would play it and the zero-sum nature of the game would be violated. So the strategy must be optimal.

Note that increasing the amount gained by winning with scissors shifted the optimal strategy to play less scissors. We often find this type of defensive maneuver in situations like this. When the rules change to benefit some specific type of action, optimal strategies often move toward counter-acting the benefited strategy instead of utilizing it. We will see that in poker, this manifests itself in that when the pot is larger, we bluff less, because a successful bluff is more profitable.

In the cases we have considered so far, all the strategy options have been equally "powerful," so to speak. Each one represented a key part of a strategy or counter-strategy, and no option was so weak that it should never be played. However, this does happen; very frequently, in fact, in more complex games. A strategy *S* is said to be ***dominated*** if there is a strategy *S'* such that $\langle S' \rangle \geq \langle S \rangle$ against all opponent strategies and $\langle S' \rangle > \langle S \rangle$ for at least one opponent strategy. *S'* is said to ***dominate S.*** (This term should not be confused with the flop game term "domination," which refers to situations where one player is reduced to three or less outs.)

This should make intuitive sense; if we have **1** strategic option, but there's some other option that does at least the same and sometimes better against all the opponent's strategies, we should always pick the better option. We can see this by looking at another variant of Roshambo. This time, instead of changing the payout matrix, we include an additional strategy option, "flower." Flower loses to each of rock and scissors and ties with paper and other flowers.

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Example 10.4 - Roshambo - F

	Player B			
Player A	Rock	Paper	Scissors	Flowers
Rock	0	-1	+1	+1
Paper	+1	0	-1	0
Scissors	-1	+1	0	+1
Flowers	-1	0	-1	0

It should be clear that no strategy that includes flower can be as good as the same strategy that substitutes paper for flower in the frequencies. Therefore, flower is dominated by paper. We also have a second terminology, ***strictly dominated,*** which means that a strategy *S' p*erforms better than *S* (not equal) against all possible opponent strategies. This often occurs when all strategic options are gone: a poker example of this is when an opponent pushes all-in preflop in a cash game (with blinds in the pot) and you are the last remaining player in the hand with AA. No matter what your opponent's strategy is, you have higher expectation from calling than from folding. Hence calling strictly dominates folding. It should be obvious that an optimal strategy cannot contain strictly dominated options: if it did, then the player could simply substitute the higher-expectation option and unilaterally improve.

Strategic options are usually valuable; in fact, we can go so far as to say that in zero-sum two - player games, ***strategic options have non-negative value.*** Adding an additional strategic option to one player's choices in a game cannot reduce his expectation in the game, because he could simply pretend that the option was not added and play his old strategy to get the old expectation. However, often he can incorporate the new option into his strategy to gain value.

One other term we frequently use to describe strategies is ***co-optimal.*** Sometimes there can be many strategies that are part of an optimal strategy pair. This can include dominated strategies in some cases. This may be surprising to some readers, but an example should clarify. Suppose that to Roshambo-F, we add yet another strategic option: "safety scissors." (Safety scissors have blunt tips and cut paper just as well as regular scissors, but other materials more poorly). Safety scissors plays just as regular scissors (i.e., loses to rock and cuts paper), but loses to flower. So in this variant, playing safety scissors is always dominated by playing regular scissors. However, strategies involving safety scissors can still be co-optimal, because they can be part of an optimal strategy pair. This is because optimal strategy pairs will never contain flower; hence, safety scissors are as good as regular scissors.

When we search for optimal strategies in poker, however, we are primarily concerned with finding undominated strategies, because picking up value from opponent's mistakes is important and occurs frequently.

When there are dominated strategic options present, we can use a recursive process to simplify the game. In the above example, we begin with the full payoff matrix:

	Player B				
Player A	Rock	Paper	Scissors	Flower	Safety
Rock	0	-1	+1	+1	-1
Paper	+1	0	-1	0	-1
Scissors	-1	+1	0	+1	0

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Flower	-1	0	-1	0	+1
Safety	-1	+1	0	-1	0

We can identify safety as a dominated strategic option by comparing the payoff values to the payoff values of scissors. In each case, the payoff value for scissors is greater than or equal to the payoff value for safety scissors. As a result, safety scissors is dominated by scissors. When we are searching for the optimal strategy, we can then reduce the game to a subgame where neither player can play safety scissors:

	Player B			
Player A	Rock	Paper	Scissors	Flowers
Rock	0	-1	+1	+1
Paper	+1	0	-1	0
Scissors	-1	+1	0	+1
Flowers	-1	0	-1	0

This matrix is, of course, Roshambo - F. We can now apply this process again, identifying Flower as the dominated strategy to remove:

	Player B		
Player A	Rock	Paper	Scissors
Rock	0	-1	+1
Paper	+1	0	-1
Scissors	-1	+1	0

And we have returned to basic Roshambo. Thus, the solution to the expanded game, including flower and safety scissors, is the same as the solution to this last sub-game. This recursive process can be used to simplify many games. This is often helpful because it reduces the number of strategic options that must be considered. To summarize:

A game G can be reduced to a sub-game G' by removing dominated strategic options from both sides. An optimal strategy pair for G' will be an optimal strategy pair for G.

The last simple game we will look at in this chapter, before going on to games directly related to poker, is called Cops and Robbers. In this game, one player is the "Cop" and the other player is the "Robber." The cop has two strategy options: to patrol or not to patrol. The robber has two strategy options as well: to attempt to rob a store, or not to. If the robber decides to rob, he wins 1 unit if the cop is not patrolling, hut loses 1 unit if the cop patrols because he gets arrested!). If the robber stays home, the cop loses 1 unit (resources, etc.) by patrolling, but nothing if he doesn't patrol. And because the game is zero-sum, the robber gains that unit when he chooses not to rob and the cop patrols.

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The payoff matrix for this game is as follows:

Example 10.5 - Cops and Robbers

	Robber	
Cop	Rob	Don’t Rob
Patrol	(1, -1)	(-1,1)
Don’t Patrol	(-1. 1)	(0, 0)

To find the optimal strategies to this game, we pick a player. Consider the cop's strategies. As before the cop will choose a percentage x of the time to patrol, and the rest of the time he will not. From Equation 1.11:

<robber, rob > = (-1)(x) + (1)(1 - x)
<robber, rob > = 1 - 2x
<robber, don't > = (l)(x) + 0(1 - x)
<robber, don't > = x

The robber is indifferent between his choices when these two values are equal.

1 - 2x = x
$x = {}^{1}/_{3}$

So the cop's optimal strategy is to patrol ${}^{1}/_{3}$ of the time.

Next we consider the robber. If he chooses a percentage *x* of the time to rob, the cop's expectations are as follows:

<cop, patrol > = (1)(x) + (-1)(1 - x)
<cop, patrol > = 2x - 1
<cop, don't > = (-1)(x) + 0(1 - x)
<cop, don't > = - x

The cop is indifferent between his choices when these values are equal.

2x - 1 = - x
$x = {}^{1}/_{3}$

So the robber's optimal strategy is to rob ${}^{1}/_{3}$ of the time.

These two values (cop patrols ${}^{1}/_{3}$ **,** and the robber robs ${}^{1}/_{3}$) are the optimal strategies for this game. We consider this game as an example of optimal strategies leading to indifference. Mixed strategies will occur if both sides have the ability to exploit each other's pure strategies. In this game, if the robber plays a pure strategy of always robbing, the cop can play a pure strategy of always patrolling. But if the cop does this, the robber can switch to a strategy of always staying home, and so on. These oscillating exploitive strategies tell us that the optimal strategies will be mixed.

This is true generally for a zero-sum two-player game; suppose two players X and Y are formulating their strategies for a game. X utilizes the pure strategy *A.* Y then exploits X by playing the pure strategy *B.* But X in turn exploits *B* by playing *C.* Y then exploits C by playing

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D. X's best response to *D* is A again. This recursive exploitation is a sign that the strategies are going to be mixed between *A* and C for X, and between *B* and *D* for Y.

If we know what components are to be mixed in the optimal strategies, we can easily solve the game systematically by writing and solving the equations that make both sides indifferent to the various mixed options. If we don't know which components are mixed - this occurs in different types of games, we often have to guess at the structure of the solution. Our guesses are sometimes called ***parameterizations,*** a topic we will revisit in discussing the solutions to [0,1] games.

In this game, however, since both the cop and the robber will be utilizing mixed strategies, we know that their expectation from each option will be equal. By setting the expectations equal to each other, we use this fact to find their strategies.

This game may seem simple and fairly unrelated to poker; but as we shall soon see, this game shares a structure and important features with a very simple poker game.

Key Concepts

- The optimal strategies to a zero-sum two-player game have the following properties: 1) As long as milted strategies are allowed optimal strategies always exist. 2) If an optimal strategy contains a mixed strategy, then the expectation of each strategic alternative must be equal against the opponent's optimal strategy.
- If a strategy-pair is optimal, neither player can improve his expectation unilaterally by changing strategies.
- An optimal strategy pair consists of two strategies that maximally exploit each other.
- Optimal strategies maximize expectation against the nemesis.
- Optimal strategies do not contain strictly dominated alternatives.
- When the players have oscillating exploitive pure strategies - that is, if X exploits Y by playing a pure strategy, Y can exploit X, after which X exploits Y again, and so on...then X and Y's strategies will be mixed.
- If a player employs a mixed strategy at any point, both of the mixed strategic options must have equal expectation against the opponent's optimal strategy.
- We can obtain simpler games that have equivalent optimal strategies by recursively removing dominated strategies from a larger game.
- In both zero-sum two-player and other types of games, strategy sets for each player that satisify the condition that no player be able to increase his expectation by acting unilaterally are called Nash equilibria. All multiplayer games with finite payout matrices have at least one such equilibrium; some games have multiple equilibria.

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